1) Cuando VA tiende a infinito

Si la tensión de Early tiende a infinito $(V_A \to \infty)$

$$r_o = \frac{V_A}{I_{CQ}} \longrightarrow \infty \quad \therefore \quad R_{OC} \longrightarrow \infty$$

LISTO, no hay que hacer más

2) REa = 0

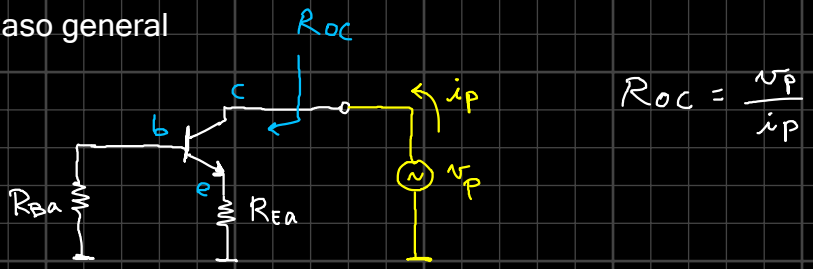
En este caso, NO circula corriente por r_π

$\rightarrow \quad R_{OC} = r_o$

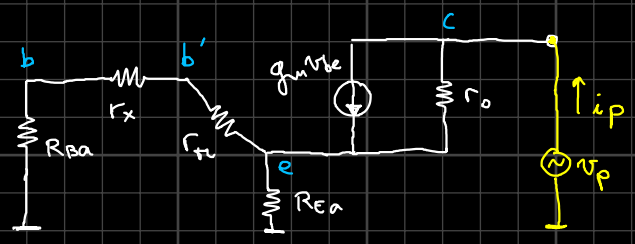


como $i_b = 0\,A \Rightarrow$ el generador no hace NADA !!

3) Caso general

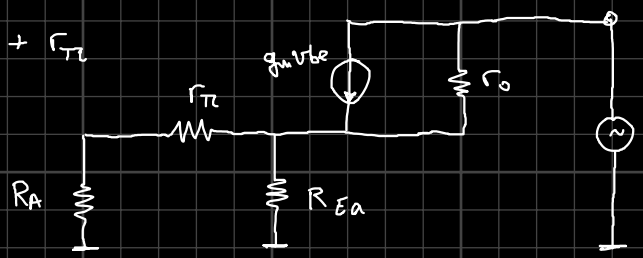


$$R_{OC} = \frac{v_p}{i_p}$$

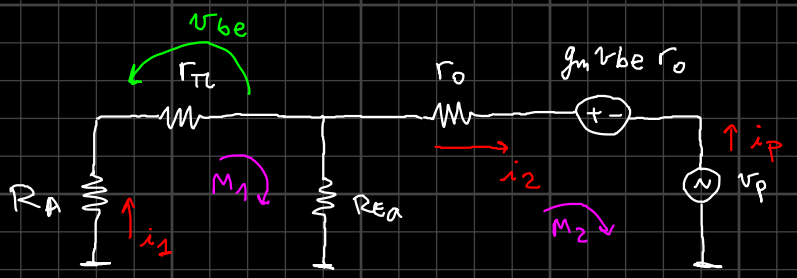


1. Asocio R para simplificar

$$R_A = R_{BA} + r_\pi$$



2. Transformo la fuente de corriente en una de tensión:



3. Planteo las mallas

M_1) $\quad 0 = i_1 (R_A + r_\pi + R_{Ea}) - i_2 R_{Ea}$

M_2) $\quad -v_p - g_m v_{be} r_o = -i_1 R_{Ea} + i_2 (r_o + R_{Ea})$

$i_2 = -i_p$

$v_{be} = i_1 r_\pi$

$R_{oc} = \frac{v_p}{i_p}$

4. Reemplazo para trabajar con i_p

M_1) $\quad i_1 = \frac{i_2 R_{Ea}}{R_A + r_\pi + R_{Ea}} = \frac{-i_p R_{Ea}}{R_A + r_\pi + R_{Ea}}$

M_2) $\quad -v_p - g_m i_1 r_\pi r_o = -i_1 R_{Ea} - i_p (r_o + R_{Ea})$

5. Reemplazo i_1 en M_2

$$-v_p + \frac{i_p R_{Ea} \cdot g_m r_\pi r_o}{R_A + r_\pi + R_{Ea}} = \frac{i_p R_{Ea}^2}{R_A + r_\pi + R_{Ea}} - i_p (r_o + R_{Ea})$$

6. Busco $R_{oc} = \frac{v_p}{i_p}$

$$-v_p = -i_p \left[\frac{-R_{Ea}^2 + R_{Ea} g_m r_\pi r_o}{R_A + r_\pi + R_{Ea}} + r_o + R_{Ea} \right]$$

$$R_{oc} = \frac{v_p}{i_p} = \frac{-R_{Ea}^2 + g_m R_{Ea} r_\pi r_o + R_{Ea} R_A + R_{Ea} r_\pi + R_{Ea}^2}{R_A + r_\pi + R_{Ea}} + r_o$$

$$R_{oc} = \frac{g_m R_{Ea} r_\pi r_o + R_{Ea} R_A + R_{Ea} r_\pi}{R_A + r_\pi + R_{Ea}} - r_o$$

$$R_{OC} = \frac{g_m R_{EA} r_\pi r_o + R_{EA}(R_{BA} + r_x + r_\pi)}{R_{BA} + r_x + r_\pi + R_{EA}} + r_o$$

Un chuetazo esta expresión, pero anda

Ejemplo: $I_{CQ} = 1\,mA$, $\beta = 200$, $V_A = 130\,V$, $r_x = 50$

$R_{BA} = 50\,k\Omega$ $R_{EA} = 10\,k\Omega$

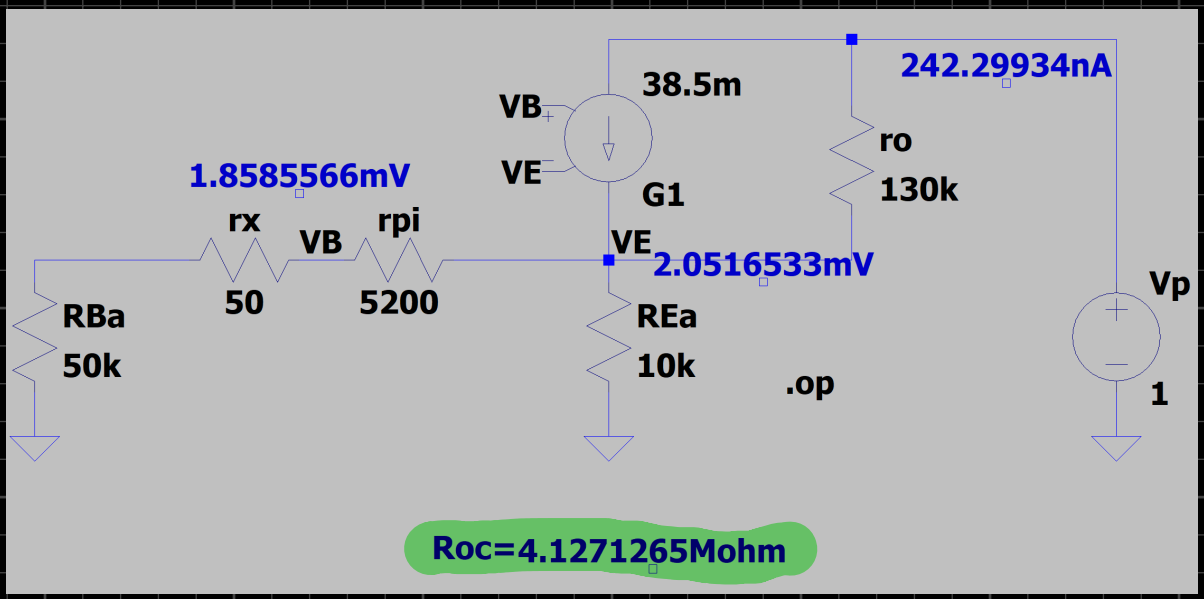
$$g_m = \frac{I_{CQ}}{V_{th}} = \frac{1}{26}\,\frac{A}{V} \approx 38,5\,\frac{mA}{V}$$

$$r_\pi = \frac{\beta}{g_m} = \frac{200}{38,5}\,k\Omega = 5k2\,\Omega$$

$$r_o = \frac{V_A}{I_{CQ}} = 130\,k\Omega$$

$$Roc = \frac{gm\ REa\ ro\ rpi + REa\ (RBa + rpi + rx)}{REa + RBa + rpi + rx} + ro$$

≈4123141.7624521074



Aproximaciones para simplificar:

I. $r_x <$ cualquier cosa, no supera los 200 Ω jamás

$$R_{OC} \approx \frac{g_m R_{Ea} r_\pi r_o + R_{Ea}(R_{Ba} + r_\pi)}{R_{Ba} + r_\pi + R_{Ea}} + r_o$$

$$R_{OC} \approx \frac{R_{Ea}}{R_{Ba} + r_\pi + R_{Ea}} \left(\overbrace{g_m r_\pi}^{\beta} r_o + R_{Ba} + r_\pi \right) + r_o$$

$$\boxed{R_{OC} \approx \frac{R_{Ea}}{R_{Ba} + r_\pi + R_{Ea}} \left(\beta r_o + R_{Ba} + r_\pi \right) + r_o}$$

$$Ros = \frac{gm\ REa\ ro\ rpi + REa\ (RBa + rpi)}{REa + RBa + rpi} + ro$$

$$\approx 4126196.319018405$$

$$Error = \frac{Roc - Ros}{Roc} \cdot 100$$

$$\approx -0.0740832293014$$

II. $\beta r_o >> R_{Ba} + r_\pi$ casi siempre

$$\Rightarrow R_{OC} \approx \frac{R_{Ea}}{R_{Ba} + r_\pi + R_{Ea}} \beta r_o + r_o = \boxed{r_o \left(1 + \frac{\beta R_{Ea}}{R_{Ba} + r_\pi + R_{Ea}} \right)}$$

$$Rop = ro \left(1 + \frac{b\ REa}{REa + RBa + rpi} \right)$$

$$\approx 4117730.061349693$$

$$Error = \frac{Roc - Rop}{Roc} \cdot 100$$

$$\approx 0.1312518805852$$

4) RBa = 0

$$R_{OC} = r_o \left[1 + \frac{\beta R_{Ea}}{0 + R_{Ea} + r_\pi} \right] = r_o \left[1 + \frac{g_m r_\pi R_{Ea}}{R_{Ea} + r_\pi} \right]$$

$R_{Ea} // r_\pi$

$$\boxed{R_{OC} \approx r_o \left[1 + g_m \left(r_\pi // R_{Ea} \right) \right]}$$